

Plane of regression for three variables

1. The equation of the plane of regression of X_1 on X_2 and X_3 is as follows:

$$X_1 = b_{12.3}X_2 + b_{13.2}X_3$$

If X_3 is considered as constant, then the graph of X_1 and X_2 is a straight line with slope $b_{12.3}$ and if X_2 is considered as constant then $b_{13.2}$ is the slope of the line.

According the principle of least squares, we have to determine the constants $b_{12.3}$ and $b_{13.2}$ in such a way that the sum of the squares of residuals is minimum i.e.

$$U = \sum (x_1 - b_{12.3}x_2 - b_{13.2}x_3)^2 = 0$$

By the principle of maxima and minima, we take the partial derivatives of U with respect to two constants: Thus,

$$\frac{\partial U}{\partial b_{12.3}} = 0 \text{ and } \frac{\partial U}{\partial b_{13.2}} = 0$$

$$\text{That's imply, } \sum 2(x_1 - b_{12.3}x_2 - b_{13.2}x_3)(-x_2) = 0 \leftrightarrow \sum (x_1 - b_{12.3}x_2 - b_{13.2}x_3)x_2 = 0$$

$$= b_{12.3} \sum x_2^2 + b_{13.2} \sum x_2x_3 - \sum x_1x_2 = 0 \quad (1)$$

Similarly, it can be found,

$$= b_{12.3} \sum x_2x_3 + b_{13.2} \sum x_3^2 - \sum x_1x_3 = 0 \text{ from } \frac{\partial U}{\partial b_{13.2}} = 0 \quad (2)$$

As we all know that,

$$\begin{aligned} \sigma_i^2 &= \frac{1}{N} \sum (x_i - \bar{x}_i)^2 \text{ for } i=1,2,3. \\ &= \frac{1}{N} \sum x_i^2 - \bar{x}_i^2 \end{aligned}$$

Since x_1, x_2, x_3 are measured from their means therefore the value means will be 0. As a result,

$$\sigma_i^2 = \frac{1}{N} \sum (x_i)^2 \text{ for } i=1,2,3. \quad (3)$$

$$\text{And } \text{cov}(x_i, x_j) = \frac{1}{N} \sum x_i x_j \text{ for } i \text{ and } j=1,2,\dots,n. \quad (4)$$

$$\text{Now, } r_{ij} = \frac{\text{Cov}(x_i, x_j)}{\sqrt{V(x_i)V(x_j)}} = \frac{\sum x_i x_j}{\sigma_i \sigma_j}$$

$$\text{It implies that } \text{cov}(x_i, x_j) = r_{ij} \sigma_i \sigma_j \quad (5)$$

Dividing N of equations (1) and (2) and using the equations (3), (4), and (5), we can write Equation (1) we can write

$$b_{12.3}\sigma_2^2 + b_{13.2}\text{cov}(x_2, x_3) - \text{cov}(x_1, x_2) = 0 \quad (6)$$

Which implies,

$$b_{12.3}\sigma_2 + b_{13.2}r_{23}\sigma_3 - r_{12}\sigma_1 = 0$$

$$\text{And from the equation (2) we can get } b_{12.3}r_{23}\sigma_2 + b_{13.2}\sigma_3 - r_{13}\sigma_1 = 0 \quad (7)$$

Where r_{12} is the total correlation coefficient of between x_1 and x_2 , r_{13} is the total correlation coefficient of between x_1 and x_3 , and r_{23} is the total correlation coefficient of between x_3 and x_2 .

Solving equations (6) and (7) we can find

$$b_{12.3} = \frac{\begin{vmatrix} r_{12}\sigma_1 & r_{23}\sigma_3 \\ r_{13}\sigma_1 & \sigma_3 \end{vmatrix}}{\begin{vmatrix} \sigma_2 & r_{23}\sigma_3 \\ r_{23}\sigma_2 & \sigma_3 \end{vmatrix}} = \frac{\sigma_1(r_{12} - r_{13}r_{23})}{\sigma_2(1 - r_{23}^2)} \quad (8)$$

And similarly we can get,

$$b_{13.2} = \frac{\sigma_1(r_{13} - r_{12}r_{23})}{\sigma_3(1 - r_{23}^2)} \quad (9)$$

If we write now

$$t = \begin{vmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{vmatrix}$$

and then $b_{12.3} = -\frac{\sigma_1 t_{12}}{\sigma_2 t_{11}}$ and $b_{13.2} = -\frac{\sigma_1 t_{13}}{\sigma_2 t_{11}}$ where t_{ij} is the cofactor of the element in the i^{th} row and j^{th} column of t matrix.

1. Finally, putting the values of $b_{12.3}$ and $b_{13.2}$ in the equation of plane regression we can find the plane of regression of X_1 on X_2 and X_3 is as follows:

$$x_1 = -\frac{\sigma_1 t_{12}}{\sigma_2 t_{11}} x_2 - \frac{\sigma_1 t_{13}}{\sigma_3 t_{11}} x_3 \rightarrow \frac{x_1}{\sigma_1} t_{11} + \frac{x_2}{\sigma_2} t_{12} + \frac{x_3}{\sigma_3} t_{13} = 0$$

Similarly, we can find the other plane of regression like:

The plane of regression of x_2 on x_1 and x_3 is

$$\frac{x_1}{\sigma_1} t_{21} + \frac{x_2}{\sigma_2} t_{22} + \frac{x_3}{\sigma_3} t_{23} = 0$$

And the plane of regression of x_3 on x_1 and x_2 is

$$\frac{x_1}{\sigma_1} t_{31} + \frac{x_2}{\sigma_2} t_{32} + \frac{x_3}{\sigma_3} t_{33} = 0$$

Problem 1

From the given data in the following table find out

1. Least square regression equation of X_1 on X_2 and X_3 .
2. Estimate the value of X_1 for $X_2 = 45$ and $X_3 = 8$.

The table is as follows:

X_1	1	2	3	4	5
X_2	3	4	5	6	7
X_3	4	5	6	7	8

Solution: Here X_1, X_2, X_3 are three random variables with their respective means $\bar{X}_1, \bar{X}_2, \bar{X}_3$

Then let us consider $x_1 = X_1 - \bar{X}_1, x_2 = X_2 - \bar{X}_2, x_3 = X_3 - \bar{X}_3$

The linear regression equation of x_1 on x_2 and x_3 is $x_1 = b_{12.3}x_2 + b_{13.2}x_3$

From the least square equation solution we have

$$b_{12.3} = \frac{\sigma_1(r_{12} - r_{13}r_{23})}{\sigma_3(1 - r_{23}^2)}$$

$$b_{13.2} = \frac{\sigma_1(r_{13} - r_{12}r_{23})}{\sigma_3(1 - r_{23}^2)}$$

The value of $\sigma_1, \sigma_2, \sigma_3, r_{12}, r_{13}, r_{23}$ can be obtained through some calculations given in the following table:

Sl. No	X_1	X_2	X_3	$x_1 = X_1 - 5$	$x_2 = X_2 - 6$	$x_3 = X_3 - 7$	$(x_1)^2$	$(x_2)^2$	$(x_3)^2$	x_1x_2	x_1x_3	x_2x_3
1	1	3	4	-4	-3	-3	16	9	9	12	12	9
2	3	5	5	-2	-1	-2	4	1	4	2	4	2
3	4	6	6	-1	0	-1	1	0	1	0	1	0
4	7	7	9	2	1	2	4	1	4	2	4	2
5	10	9	11	5	3	4	25	9	16	15	20	12
Total	25	30	35	0	0	0	50	20	34	31	41	25

$$\text{Here, } \bar{X}_1 = \frac{25}{5} = 5, \bar{X}_2 = \frac{30}{5} = 6, \bar{X}_3 = \frac{35}{5} = 7$$

$$\text{And } \sigma_1^2 = \frac{1}{5} \sum x_i^2 = \frac{50}{5} = 10 \rightarrow \sigma_1 = \sqrt{10} = 3.162$$

$$\sigma_2^2 = \frac{1}{5} \sum x_2^2 = \frac{20}{5} = 4 \rightarrow \sigma_2 = 2$$

$$\sigma_3^2 = \frac{1}{5} \sum x_3^2 = \frac{34}{5} = 6.8 \rightarrow \sigma_3 = \sqrt{6.8} = 2.608$$

$$r_{12} = \frac{\sum x_1 x_2}{N \sigma_1 \sigma_2} = \frac{31}{5 * 3.162 * 2} = 0.98$$

$$r_{13} = \frac{41}{5 * 2.608 * 3.162} = 0.994$$

$$r_{23} = \frac{25}{5 * 2.608 * 2} = 0.9596$$

Hence, from the formula we can get

$$b_{12.3} = 0.527 \text{ and } b_{13.2} = 0.818$$

The equation of regression plane of x_1 on x_2 and x_3 is $x_1 = 0.527x_2 + 0.818x_3$

After substituting the value we can get $X_1 = -3.891 + 0.527X_2 + 0.818X_3$

For 2nd part of the problem we have to put the values of the random variables, we get,
 $X_1 = 26.368$

Assignment:

From the given data in the following table find out

1. Least square regression equation of X_1 on X_2 and X_3 .
2. Estimate the value of X_1 for $X_2 = 6$ and $X_3 = 8$.

The table is as follows:

X_1	2	6	8	10
X_1	4	5	9	12
X_1	4	6	10	12

